AI-Based Smart Retail Recommendation System for Apparel

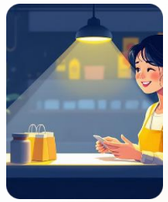


Problem Definition



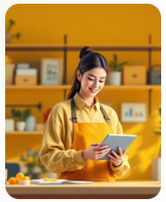
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Users & Stakeholders



Retail Customers

Seek discovery, relevance and a frictionless in-store journey.



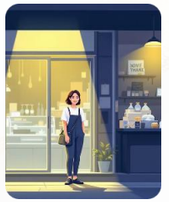
Store Managers

Need actionable insights to optimise layout, staff and stock in real time.



Marketing Teams

Require behaviour-driven signals to tailor promotions and measure campaign impact.



Business & Supply Chain

Care about inventory flow, cost control and reliable demand forecasts.



Overall Problem

- Traditional retail systems only handle basic tasks like **billing and inventory**
- **Retailers don't know how customers behave**, feel, or interact with products in-store
- Huge amounts of data from **CCTV and POS systems go unused**
- **Hard to make good decisions** without real insights

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Existing Solutions — What Retailers Use Today

Basic CCTV Monitoring

Captures video but requires manual review and adds limited behavioural context.

Traditional Inventory Systems

Track stock levels but often operate on delayed data and siloed processes.

E-commerce Recommendation Engines

Powerful online, but not adapted to in-store interactions or real-time context.

Standard Marketing Analytics

Provide campaign summaries but lack linkages to in-store behaviour and emotions.



Limitations of Current Systems

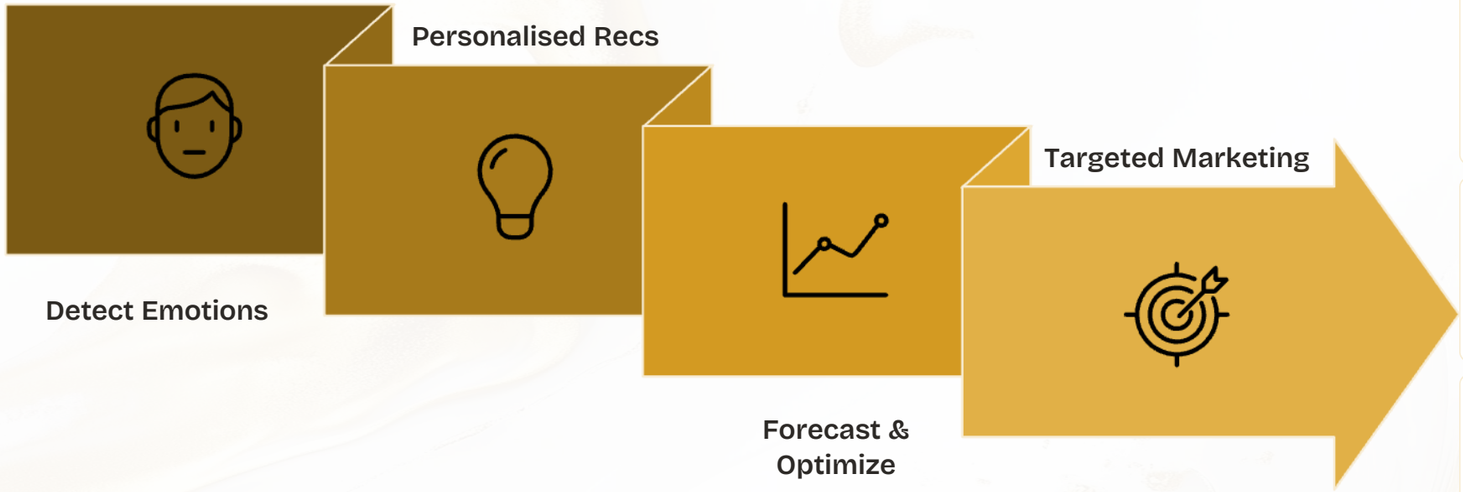
- **Single-aspect analysis** — systems focus on **one data stream** (e.g., CCTV or inventory) rather than integrated signals.
- **Absence of real-time behavioural and emotional insight** — delays prevent timely actions.
- **Poor integration between customer behaviour,** inventory and marketing — **decisions are fragmented**.
- **Inability to perceive in-store customer emotions** — critical signals for personalization are missing.

How Our System Solves the Problem

- **Understand customer emotions** and shopping behavior in real time
- **Improve product recommendations** and **shopping experience**
- **Reduce stock shortage**s and **overstock problems**
- **Design smarter, more effective marketing campaigns**
- Turns traditional retail into a data-driven, **AI-powered environment**



Proposed Solution: AI-Driven Smart Retail Intelligence



Real-time sensing feeds recommendations, forecasting and campaign engines — producing actionable insights and automated responses for staff, displays and inventory systems.

Real-time Emotion & Behaviour Analysis

Detects engagement, frustration, dwell time and interactions at product level.

AI-Powered Recommendations

Smart mirrors and shelf displays provide contextual, timely product suggestions to shoppers.

Demand Forecasting & Inventory Optimisation

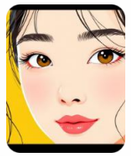
Short-term and seasonal forecasts align stock flow with observed in-store demand signals.

Personalised Marketing Intelligence

Precision campaigns triggered by in-store behaviour, improving engagement and conversion.

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System Components — Detailed Overview



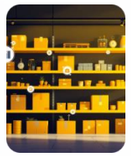
1. Customer Emotion & Behaviour Detection

Edge AI analyses anonymised facial cues, posture and gaze to infer engagement, frustration and interest while preserving privacy.



2. Smart Mirror Recommendation System

Contextual, in-moment suggestions combining visual fit, stock availability and historical preferences.



3. Smart Inventory & Stock Flow Optimisation

Demand signals inform replenishment, allocation and redistribution to reduce stockouts and overstocks.



4. Personalized Marketing Intelligence

Targeted promotions and tests driven by in-store behaviour, improving and customer relevance.

Together these components create a closed-loop system: **sense → predict → act → learn**. The result is better customer experiences, more efficient operations and measurable commercial uplift.





Customer Emotions and Behavior Detection

Smart Clothing Retail Recommendation System

A real-time AI system that combines emotion detection with behavior tracking to automatically notify store staff when customers need assistance.

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Problem Definition & Research Gap

Current Limitations

Many retail systems rely solely on past purchase data and fail to understand customer emotions in real time. Shoppers may feel confused or need help but have no way to signal it.

Research Gap

Existing systems can detect emotion OR behavior, but most only generate reports after the fact. They do not provide real-time assistance to customers while shopping.

Novelty

We combine emotion detection with behavior tracking to detect customer difficulties in real time and automatically notify store staff to provide immediate help.

Objectives

Main Objective

Develop an AI-based system to detect customer emotions and behavior in real time and notify store staff when assistance is needed.

Specific Objectives

- Detect facial emotions using computer vision
- Track customer movement and product interaction
- Identify customers facing shopping difficulties
- Send automatic alerts to store staff





Methodology, Data & Evaluation

01

Design Science Research

Iterative development: build, evaluate with stakeholders, refine the system for in-store deployment.

02

Tools & Platforms

Python, OpenCV for vision pipelines; TensorFlow / PyTorch for model training and inference; lightweight edge deployment options.

03

Data & Ethics

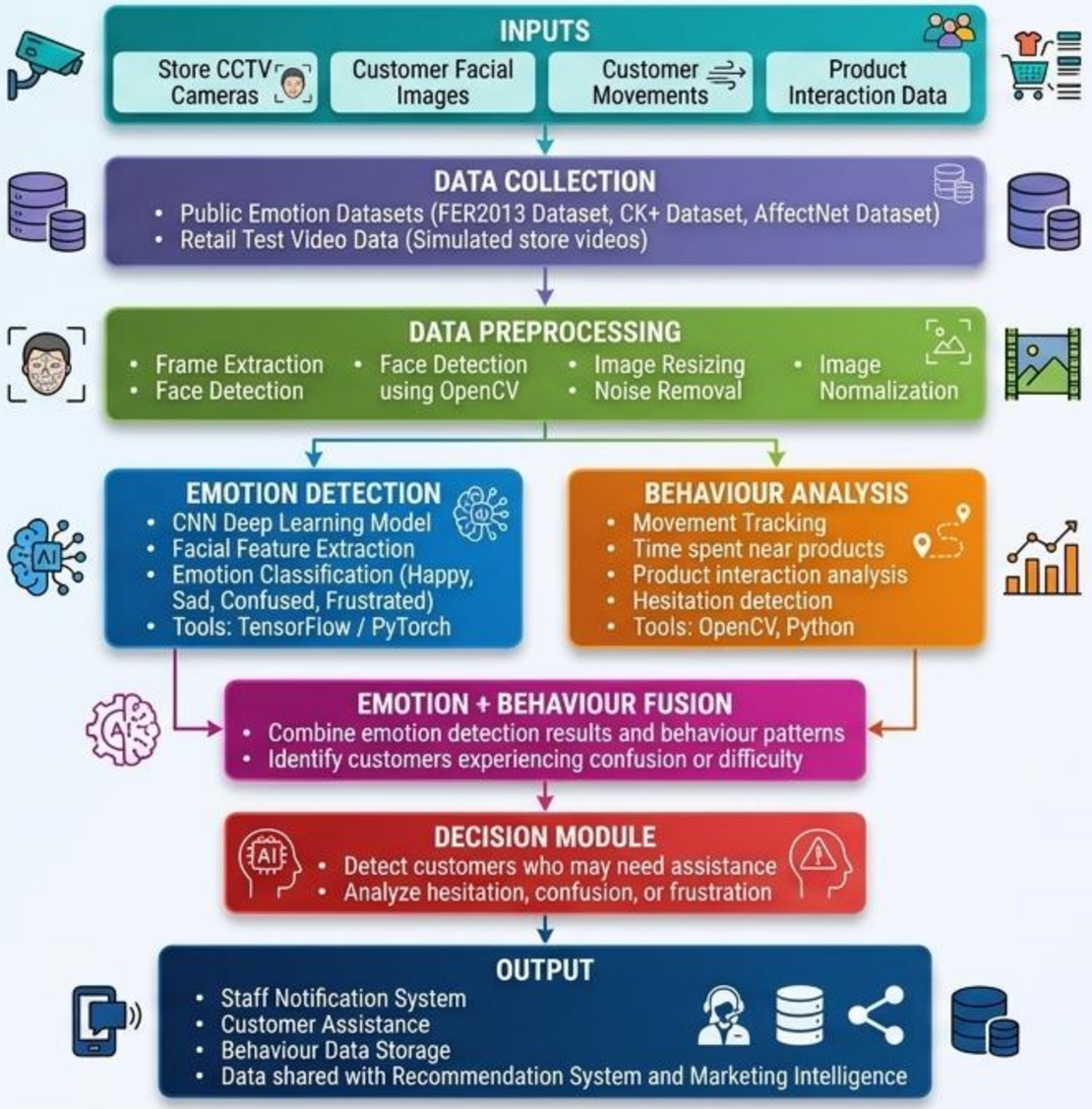
Use public facial-emotion datasets and anonymised retail video for testing; do not store personally identifiable information — only transient features and timestamped events.

04

Validation Metrics

Evaluate using accuracy, precision, recall, F1 score for detection tasks, plus system response time and staff response rate in realTrials.

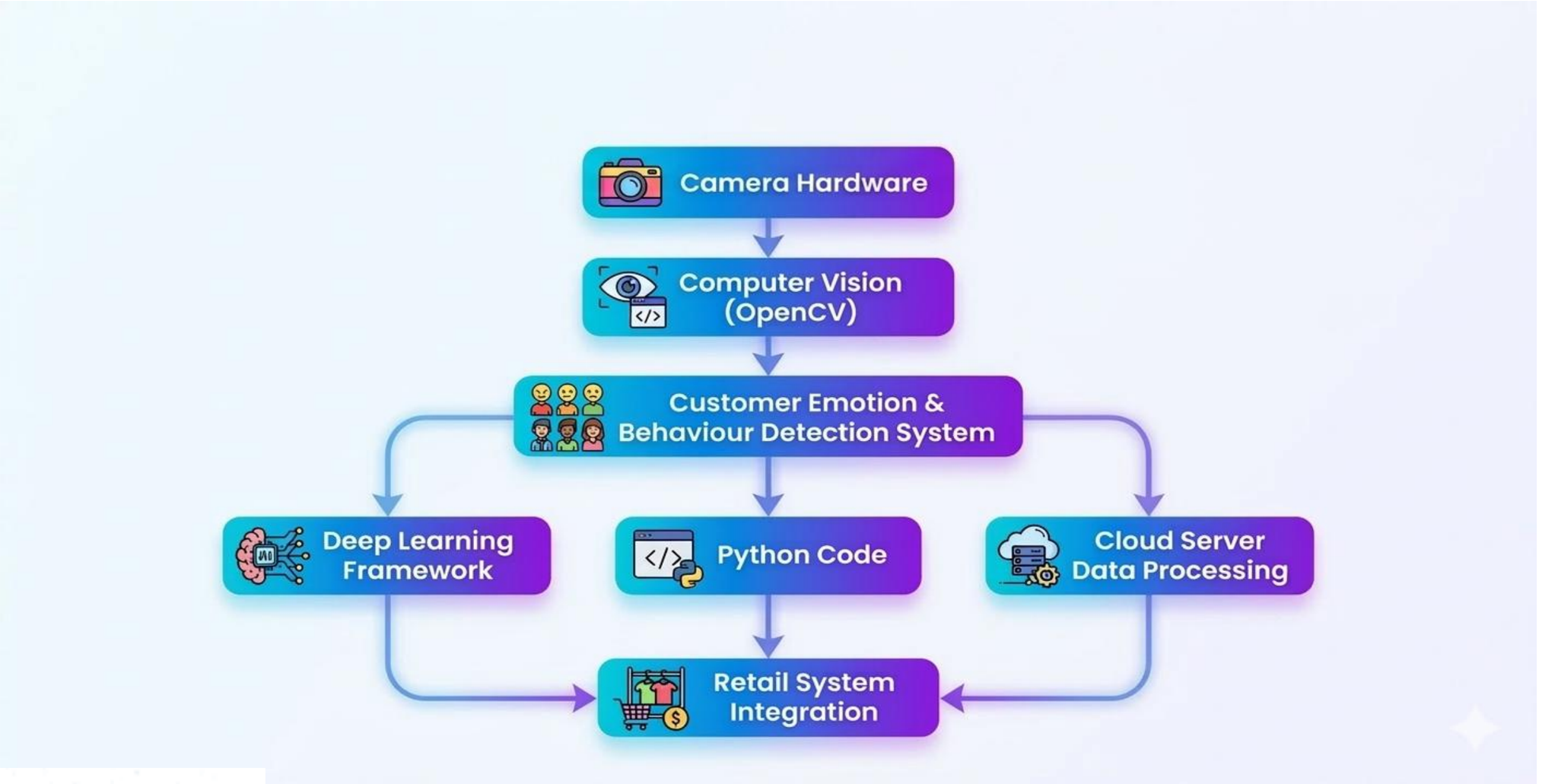
Customer Emotions and Behaviour Detection in Smart Retail Recommendation System



Technologies Used

- Computer Vision (OpenCV) – Used to detect faces and process customer images from store cameras.
- Deep Learning (CNN) – Used to recognize customer emotions from facial expressions.
- TensorFlow / PyTorch – Frameworks used to train and run the emotion detection model.
- Python Programming Language – Used to develop the system and implement algorithms.
- Camera Sensors (CCTV) – Capture real-time customer facial expressions and movements.
- Cloud / Local Server – Used for data processing and storing behaviour analysis results.

Dependency Architecture Diagram



Implementation Overview

Smart Retail Recommendation System

- Camera Input – Store cameras capture customer faces and movements.
- Face Detection – OpenCV detects faces from video frames.
- Emotion Detection – CNN deep learning model identifies emotions (happy, confused, etc.).
- Behaviour Tracking – Tracks customer movement and time near clothing products.
- Analysis – Combines emotion and behaviour data to detect customer difficulty.
- Alert System – Sends notification to store staff for assistance.







Smart Mirror Recommendation System

Smart Clothing Retail Recommendation System

Current in-store smart mirrors focus on virtual try-on but stop at visuals.
They rarely provide personalized product recommendations tied to live inventory or interpret customer emotions and interactions.

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Problem & Research Gap

Problem Definition

- Smart mirrors in retail stores are mainly used for virtual clothing try-on.
- Most systems only display how clothes look on customers.
- They do not provide intelligent product recommendations.

Research Gap

- Lack of personalization in recommendations.
- Customer emotions and behaviour are not analysed.
- Smart mirrors are not connected with real-time store inventory.

Our Novelty

- AI-powered smart mirror recommendation system.
- Analyzes customer facial expressions and interaction behavior.
- Recommends suitable clothing items based on available store inventory.

Objectives

Main Objective

Develop an AI-based smart mirror system that analyzes customer features and interaction behavior to recommend suitable clothing products using store inventory data.

Specific Objectives

- Develop a smart mirror interface with virtual try-on.
- Detect customer facial features and interaction behaviour using computer vision.
- Generate AI-based clothing recommendations.
- Connect recommendations with real-time store inventory.



Technologies Used

- Computer Vision – Detect facial features and customer interaction.
- Convolutional Neural Networks (CNN) – Analyse facial expressions and emotional states.
- Active shape Models(ASM) – Detect facial landmarks for feature analysis.
- FMCW Rader Sensing – Detect physiological emotional signals without collecting sensitive personal data.
- Affective Edge Computing(AEC) - Process data locally for faster response and privacy protection.
- Federated Learning (FL) – Train AI models securely without sharing row customer data.

Research Methodology

01

Research Approach

Design science research approach for developing a technological for smart retail recommendations.

02

Tools & Platforms

Python, OpenCV, TensorFlow/PyTorch , MySQL/ Firebase for implementation

03

Data & Ethics

Public facial emotion datasets and retail video data for testing. No personal identity data stored.

04

Validation Metrics

Accuracy, precision, recall, F1 score, and response time evaluation

INPUTS

- Customer from Camera
- Customer Facial Expressions
- Customer Body Posture
- Customer Interaction Behaviour
- Product Database
- Store Inventory Data

1. IMAGE CAPTURE & PREPROCESSING

- Camera captures customer image in front of mirror
- Image preprocessing using OpenCV
- Noise removal
- Image normalization
- Face detection

2. FEATURE DETECTION MODULE (Computer Vision)

- Facial feature detection
- Body posture detection
- Body posture detection
- Interaction behaviour analysis
- Interaction behaviour analysis
- Facial landmark detection (ASM)

3. EMOTION & BEHAVIOUR ANALYSIS

- Deep learning model (CNN)
- Facial expression analysis
- Detect emotional state
- Identify customer preferences

4. AI RECOMMENDATION ENGINE

- Recommendation algorithm
- Match detected features with clothing styles
- Identify suitable clothing products

5. INVENTORY VALIDATION MODULE

- Database connection (MySQL / Firebase)
- Check product availability
- Filter unavailable products
- Select available clothing items

6. SMART MIRROR INTERFACE



- Display recommendations on smart mirror
- Virtual clothing try-on
- Product details (price, size, color)
- Interactive mirror interface

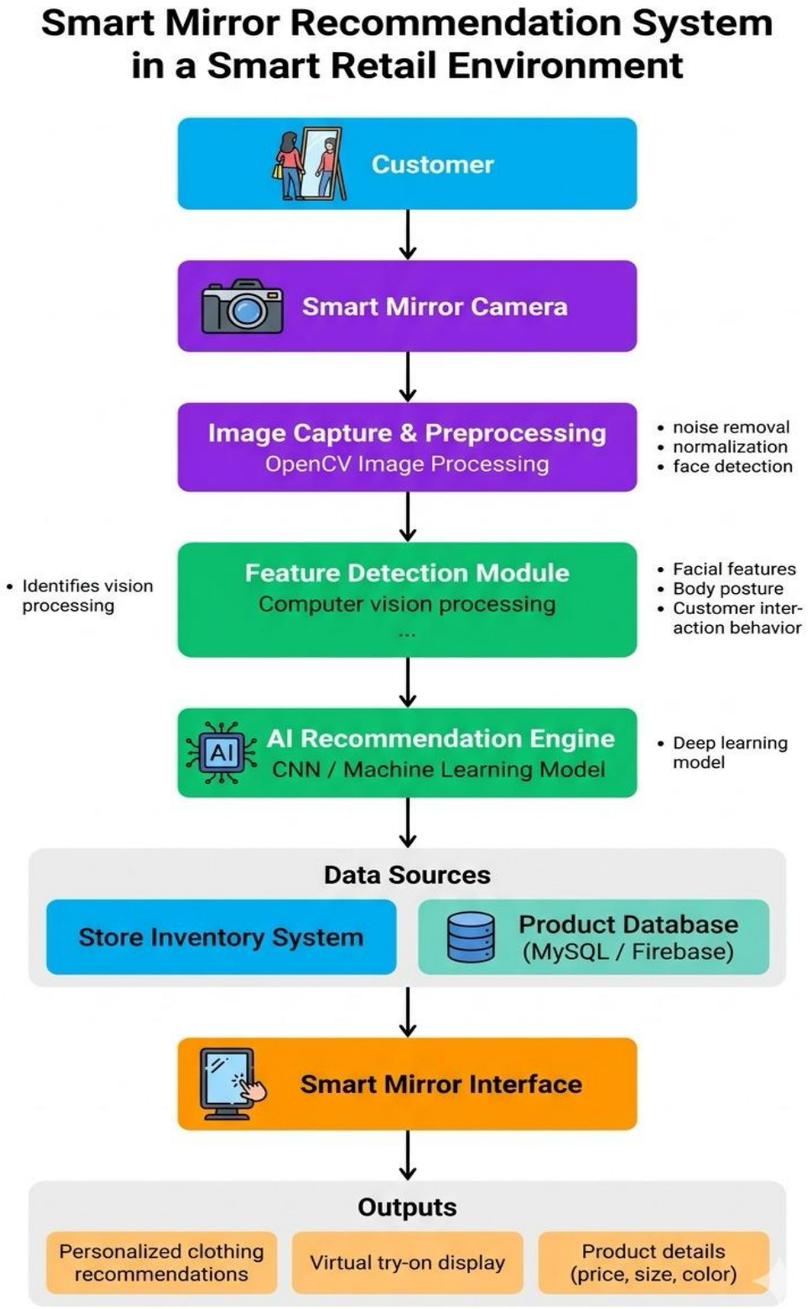
OUTPUTS

- Personalized clothing recommendations
- Virtual try-on display
- Available product suggestions
- Improved shopping experience

Implementation Overview

Smart Retail Recommendation System

- The Smart Mirror Recommendation System lets customers try clothes virtually using a digital mirror
- The System captures the customer's image using camera.
- Computer vision analyes the customer's facial expressions and behaviour.
- An AI recommendation system suggests suitable clothing items.
- The system also connects with the store inventory to recommend products that are available in the store.



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SMART MIRROR RECOMMENDATION SYSTEM
(Main Central System)

CAMERA MODULE



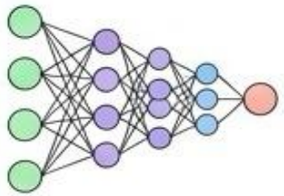
Image Capture



OpenCV

Capture customer images from the smart mirror camera.

AI / COMPUTER VISION MODULE



CNN Model



Feature Analysis

Analyze facial features, body posture, and customer behavior.

DATABASE MODULE

	Name	Product	Details
1	Slot	Deals	Wibite
2	Product	Products	Elim
3	Product	Barrum	Natas
4	Product	Msals	Green
5	Size	Fend	Stars

Product Data

MySQL / Firebase

Store product information and inventory data.



Smart Inventory & Stock Flow Optimization



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Contributions & Novelties

AI-based demand forecasting

Predict demand per store and SKU

Improved inventory management

Synchronise stock across retail network

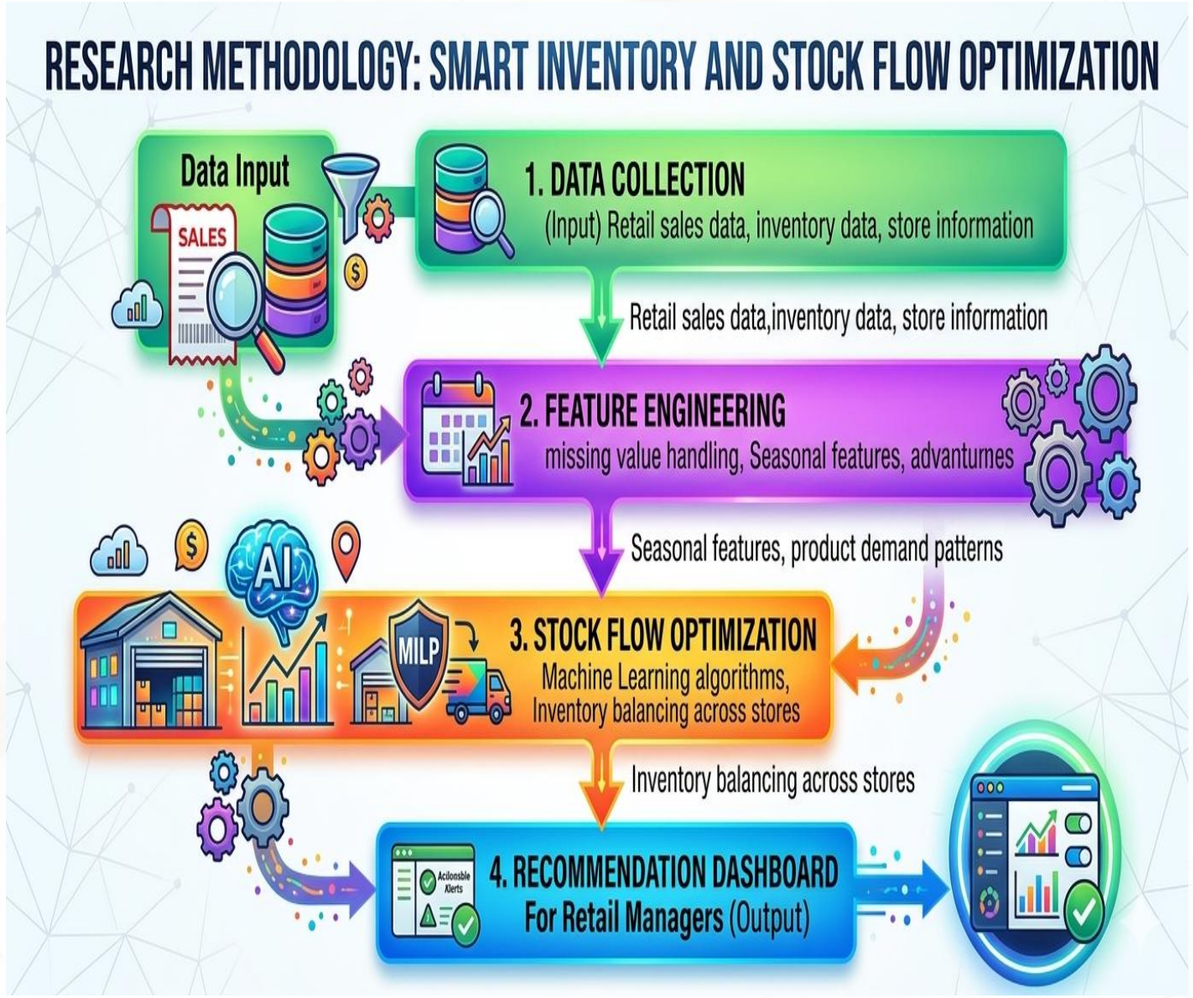
Optimised stock distribution

Reduce transit and holding costs

Less shortages & excess

Maintain target service levels

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Problem, Objective & Sub-objectives

Problem Definition

- Shortages at high-demand stores
- Excess stock at low-demand locations
- Inefficient inter-site transfers

Research Objective

- Develop AI system for demand prediction
- Optimise stock distribution decisions

Sub-objectives

- Analyse historical sales data
- Apply ML for demand forecasts
- Optimise transfers between locations
- Provide explainable insights

Domain Knowledge & Methodology

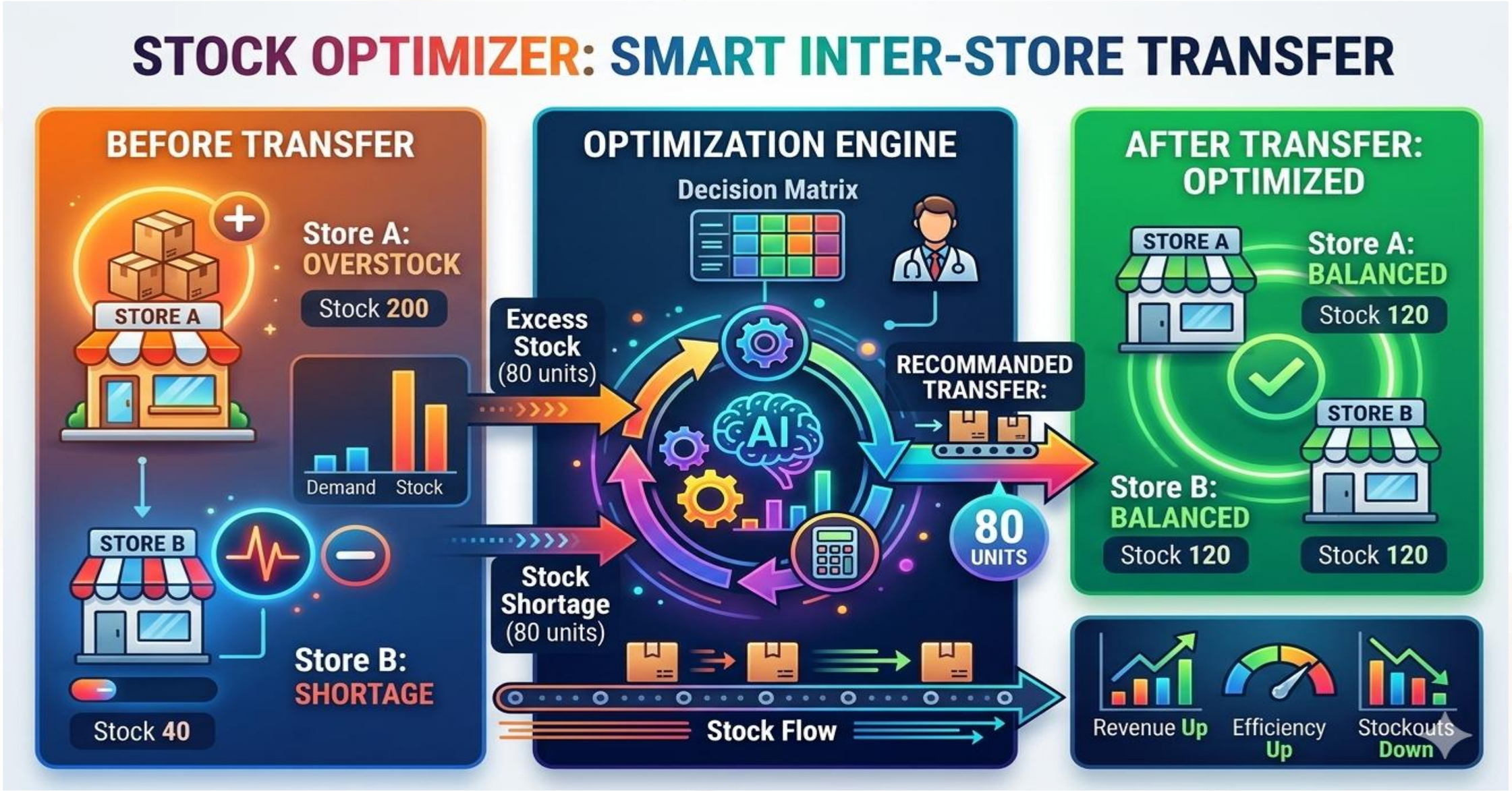
Application Areas

- Retail inventory management
- AI & machine learning
- Sales data analytics

Research Methodology

- Data collection (POS, stock logs)
- Data preprocessing & feature engineering
- Machine learning forecasting models
- Mathematical optimisation for stock flow




STOCK OPTIMIZER: SMART INTER-STORE TRANSFER
BEFORE TRANSFER
STORE A
Store A:
OVERSTOCK
Stock 200
Demand
Stock
STORE B
Store B:
SHORTAGE
Stock 40
OPTIMIZATION ENGINE
Decision Matrix
Excess Stock
(80 units)
AI
RECOMMANDED TRANSFER:
80
UNITS
Stock Shortage
(80 units)
Stock Flow
AFTER TRANSFER:
OPTIMIZED
STORE A
Store A:
BALANCED
Stock 120
STORE B
Store B:
BALANCED
Stock 120
Stock 120
Revenue Up
Efficiency
Up
Stockouts
Down

Technologies, Validation & Ethics



Technologies

- **Machine Learning:** Random Forest (demand forecasting) using Python & Scikit-learn
- **Explainable AI:** SHAP / feature importance analysis to interpret predictions
- **Data Processing & Analytics:** Pandas, NumPy for data analysis and feature engineering
- **Optimization:** Google OR-Tools or SciPy for stock flow optimization
- **Visualization:** Power BI / dashboard for inventory insights

Validation

- Forecast Accuracy: Mean Absolute Error (MAE), Root Mean Square Error (RMSE)
- Inventory Efficiency: Reduction of stock shortages and excess inventory
- Baseline Comparison: Compare AI predictions with traditional forecasting methods

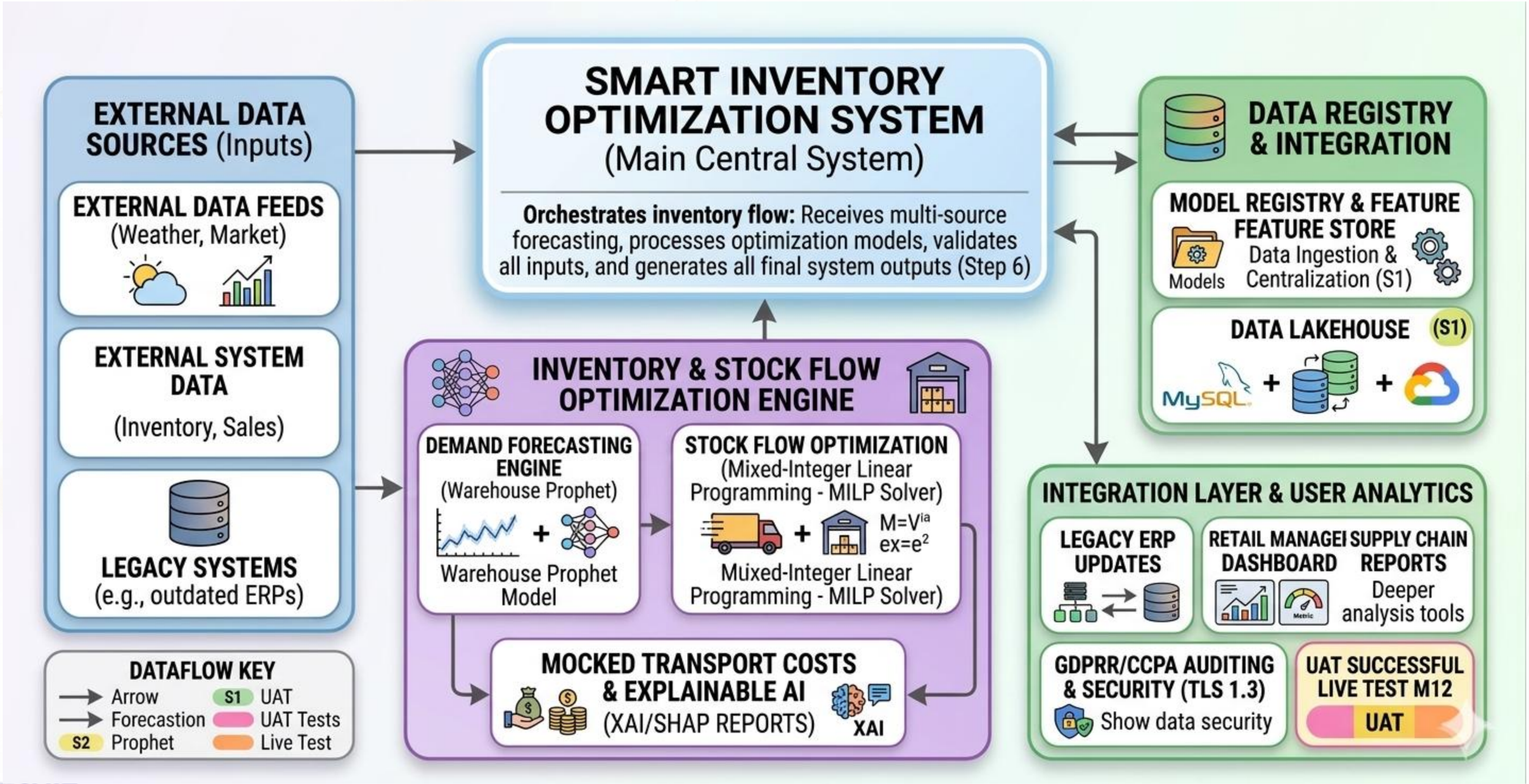
Data & Ethics

- Dataset: Simulated retail sales dataset including product, store, and inventory data
- Data Privacy: No personal or customer data used in the study
- Ethical Use: AI outputs are explainable and support human decision-making

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System Overview & Dependencies



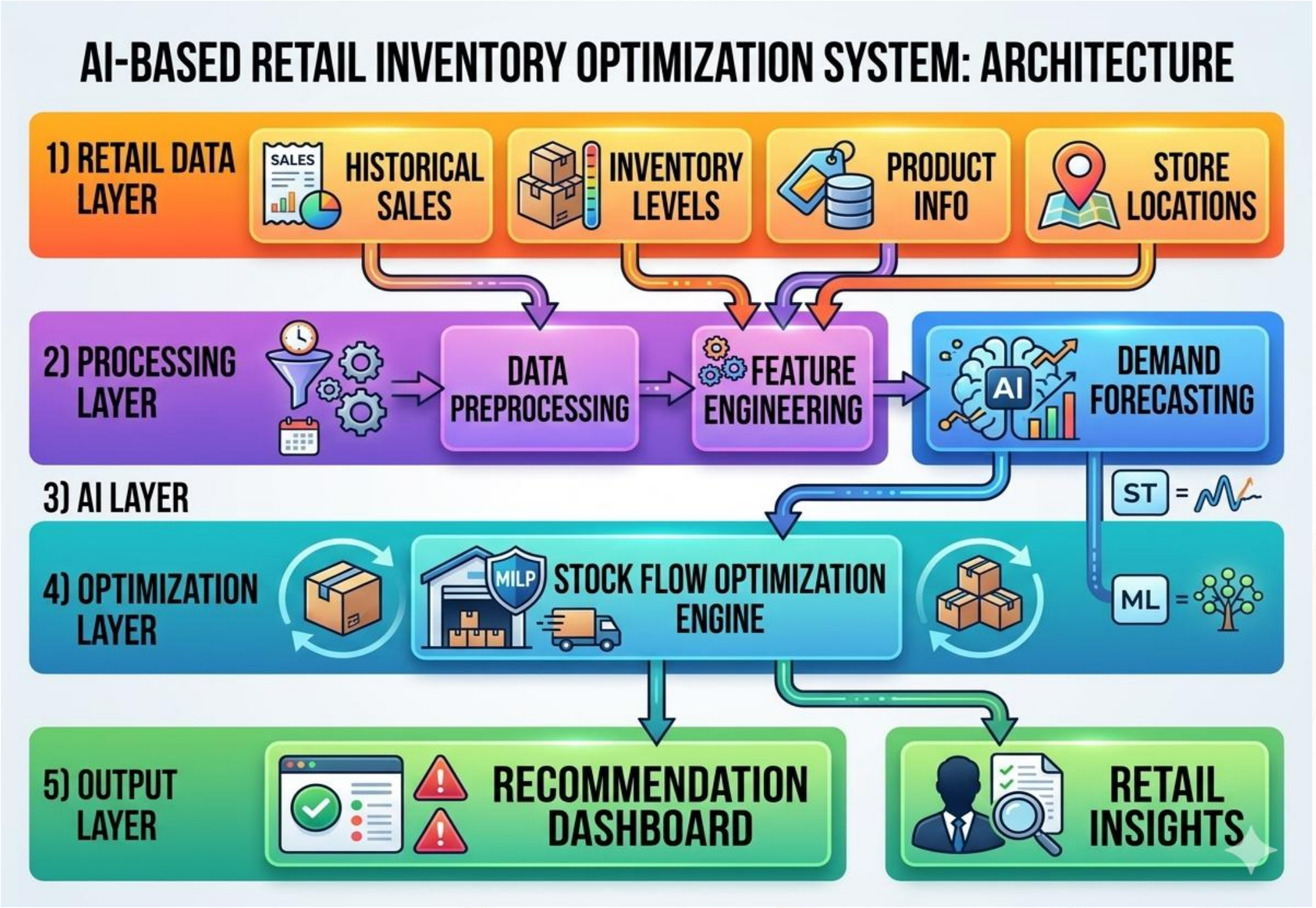
System Overview & Dependencies

System Overview

- Sales Data → AI Demand Forecasting
- Forecasts → Stock Optimisation
- Optimiser → Inventory Recommendations

Key Dependencies

- Retail sales database
- Real-time inventory monitoring
- AI forecasting module
- Decision-support dashboard



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Personalized Marketing Intelligence

Contribution : Marketing Intelligence Dashboard

An intelligence marketing dashboard which analyses customer emotions , behaviors and help stakeholders to gain insight for better decision making

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Personalized Marketing Intelligence | Problem Definition

Retail businesses often experience challenges in delivering effective marketing promotions:



Generic promotions



Lack of insights into promotion performance and customer responses



Low customer engagement due to poorly targeted promotions



Limited understanding of individual customer preferences and purchasing behavior



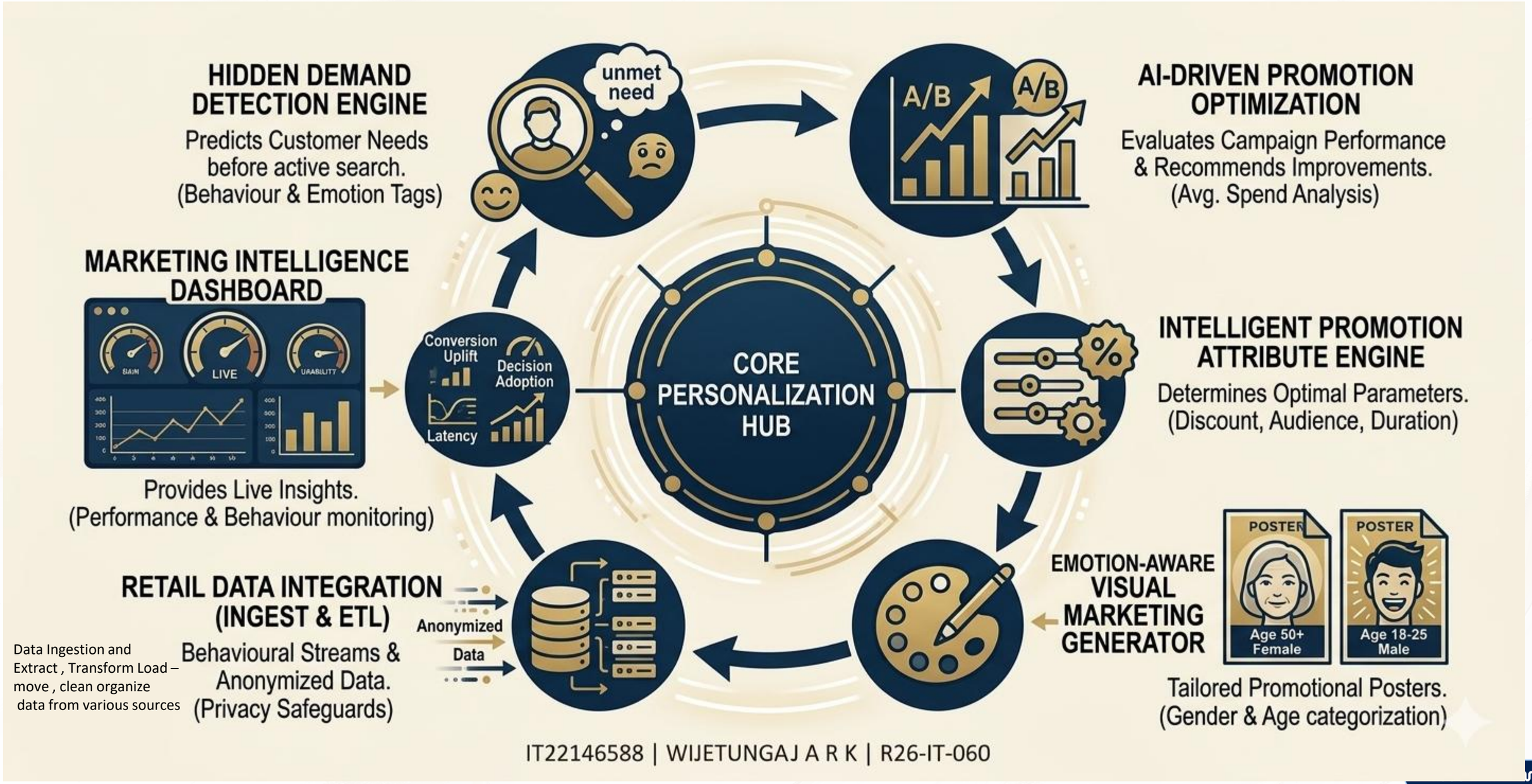
Inability to identify hidden or future customer demand



Manual marketing campaign creation which reduces efficiency

Personalized Marketing Intelligence

Novelties :



Data Ingestion and Extract , Transform Load – move , clean organize data from various sources



Research Objectives

Problems: generic promotions; inability to detect hidden or future customer demand; low engagement from poorly targeted promotions; manual campaign creation; lack of real-time promotion insights.

Research Objective

Develop an **AI-powered system that analyses behaviour, predicts hidden demand, and generates optimised personalized promotions**

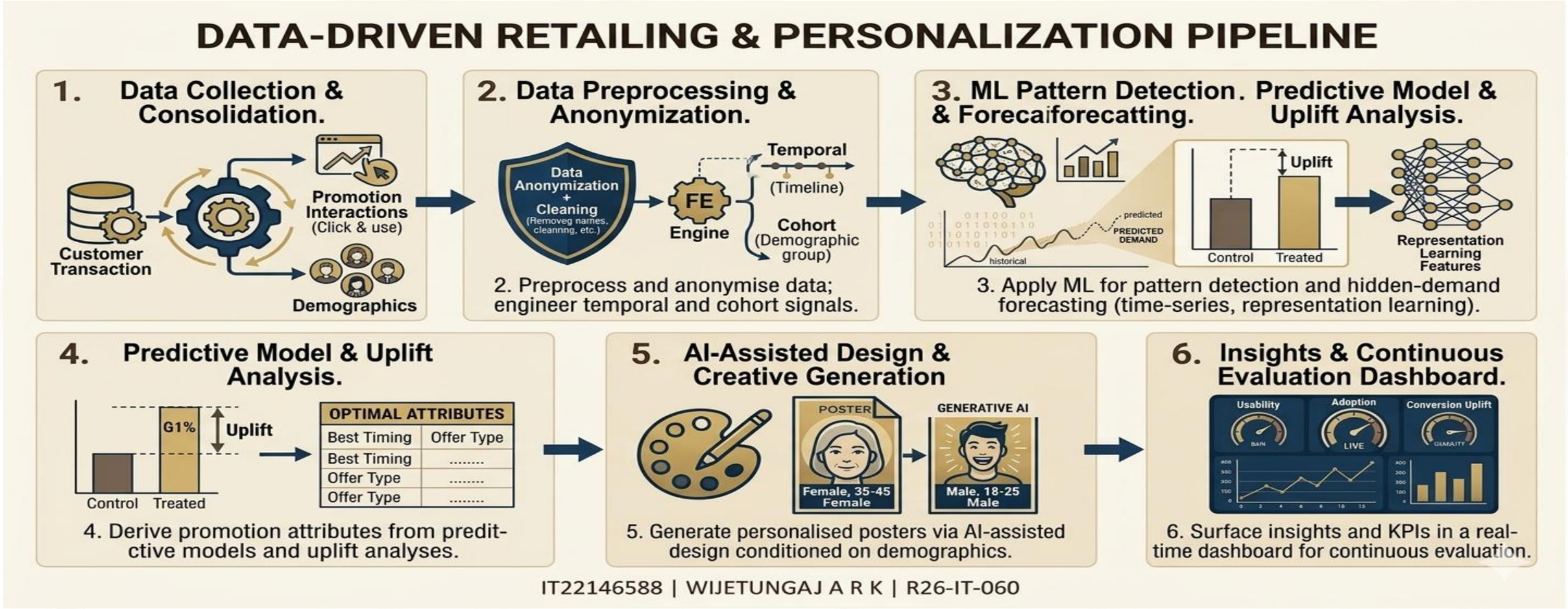
Goal : To raise engagement and retail sales.

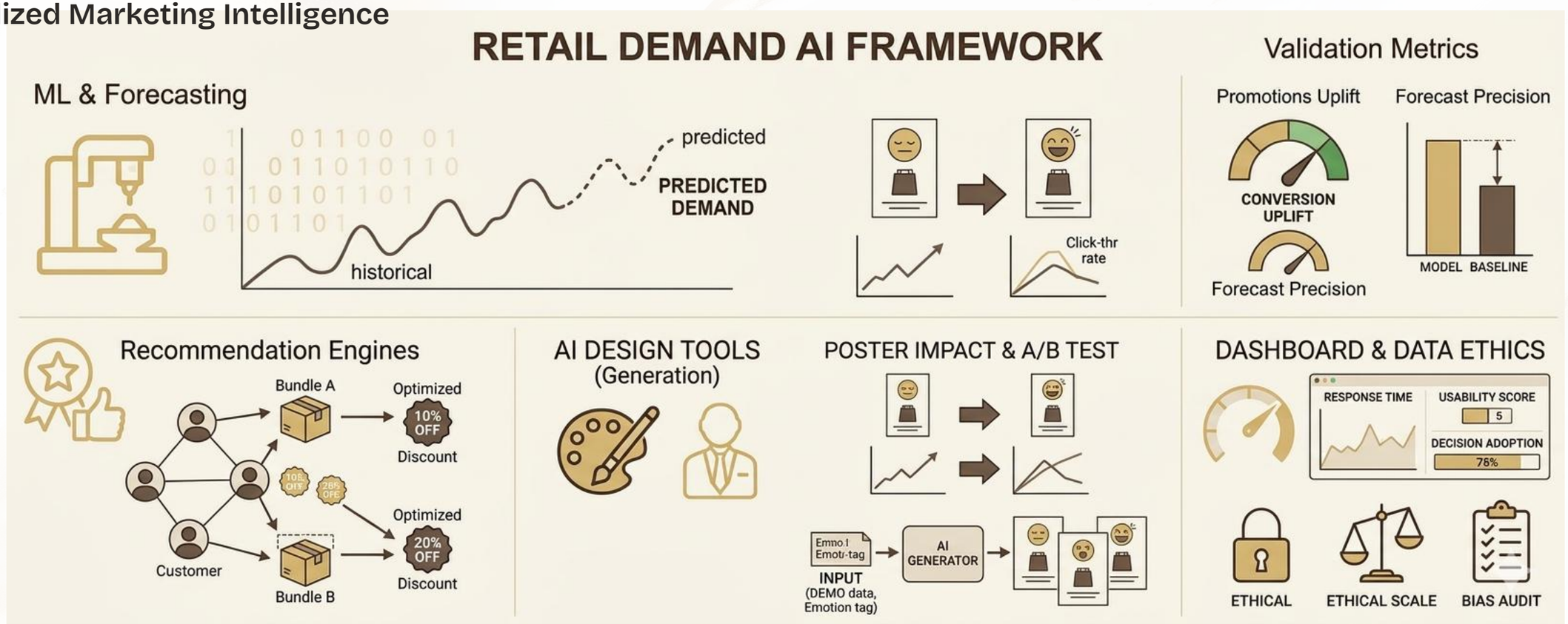
Key Sub-objectives

- Analyse **purchasing and promotion interaction data**
- **Detect hidden demand with machine learning**
- Generate optimal promotion **attributes** (discount, audience, timing)
- Auto-create demographic-specific marketing posters
- Deliver a real-time performance dashboard

Application of Domain Knowledge & Methodology

Domain knowledge spans retail marketing, customer behaviour analysis, machine learning segmentation, transaction analytics and marketing design intelligence for emotion-aware creative outputs.





ML & Forecasting

Time-series models, representation learning, anomaly detection for hidden-demand prediction.

Analytics & Visualisation

Real-time streaming analytics, KPI visualisation, and anomaly alerts for campaign monitoring.

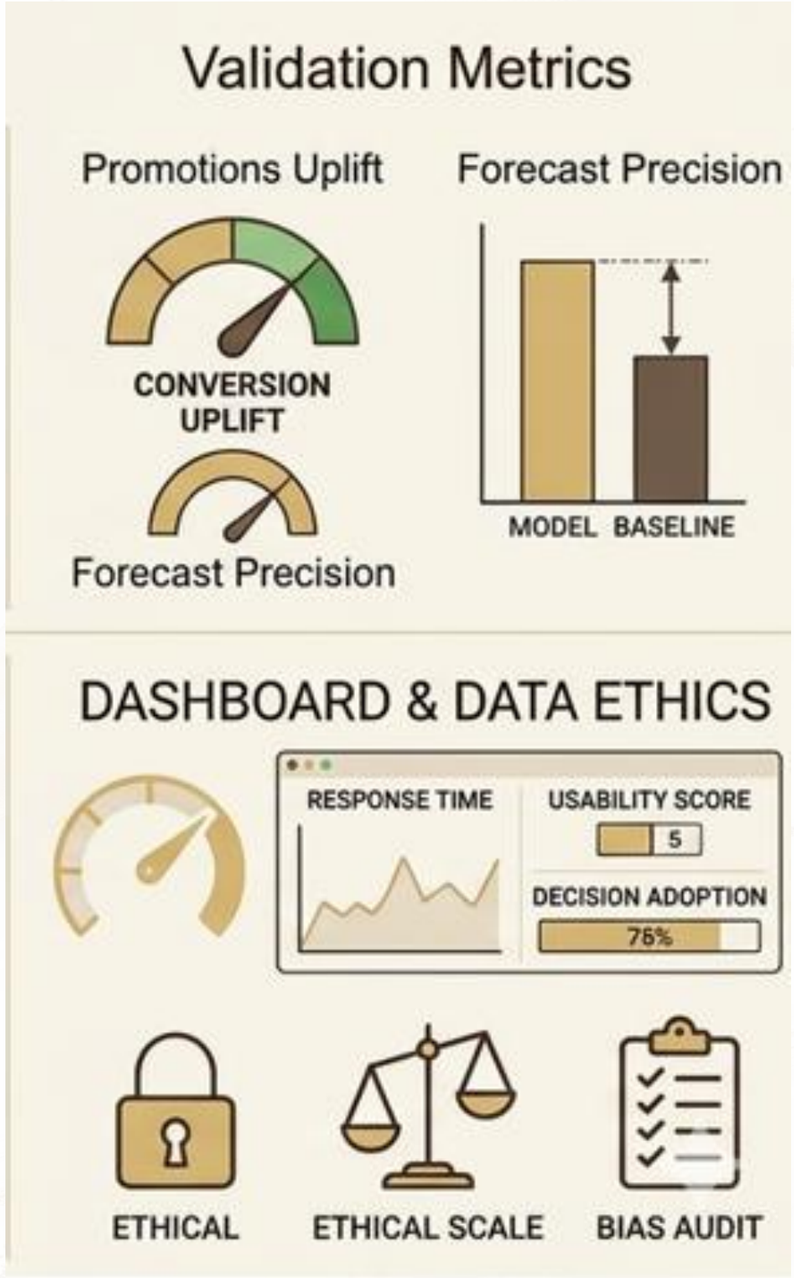
Recommendation Engines

Association mining and contextual bandits for personalised bundles and discount optimisation.

AI Design Tools

Conditional generative models to produce emotion-aware promotional posters tailored by demographics.

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Validation Metrics

- **Hidden-demand prediction**: precision, recall, and lift over baseline.
- **Promotion effectiveness**: engagement rate, conversion uplift, revenue per promotion.
- **Poster impact**: click-through and creative A/B test results by demographic cohort.
- **Dashboard performance**: latency, usability, accuracy and decision adoption metrics.

Data & Ethics

Use simulated or anonymised datasets for model training; strict adherence to data privacy guidelines; data usage restricted to academic research; bias audits and fairness checks applied to model outputs.

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System Overview & Dependencies

End-to-end pipeline: Customer Data → AI Behaviour Analysis → Hidden Demand Detection → Promotion Optimization → Personalized Poster Generation → Real-Time Dashboard.

Core System Components



Retail Customer Transaction Database
(ingest & ETL)

AI Behaviour & Demand Prediction Module
AI model for consumer patterns





Promotion Attribute Optimisation Engine
Finding optimal offer settings

Poster Generation & Design Module (emotion-aware)
Creative and layout engine





Marketing Campaign Manager & Delivery System
Campaign execution control

Real-Time Analytics & Visualisation Dashboard
System monitoring & reporting



Operational Dependencies & Next Steps

- Integrate anonymised transaction streams and promotion logs.
- Deploy lightweight inference service for live prediction and campaign scoring.
- Run controlled field experiments to validate uplift and iterate creative models.
- Establish governance: privacy safeguards, fairness audits, and reproducible research artifacts.



Integrate anonymised transaction streams and promotion logs.

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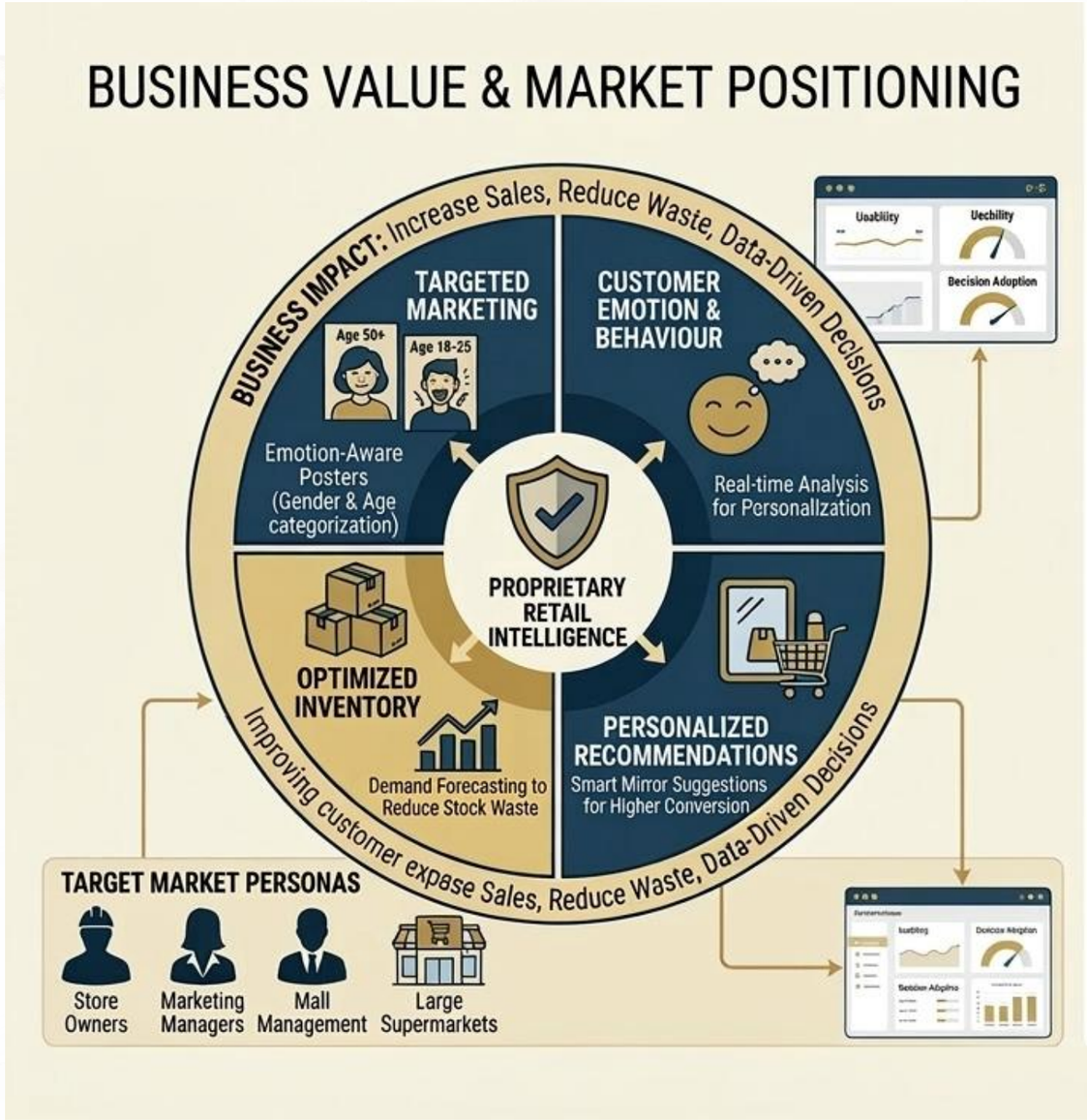
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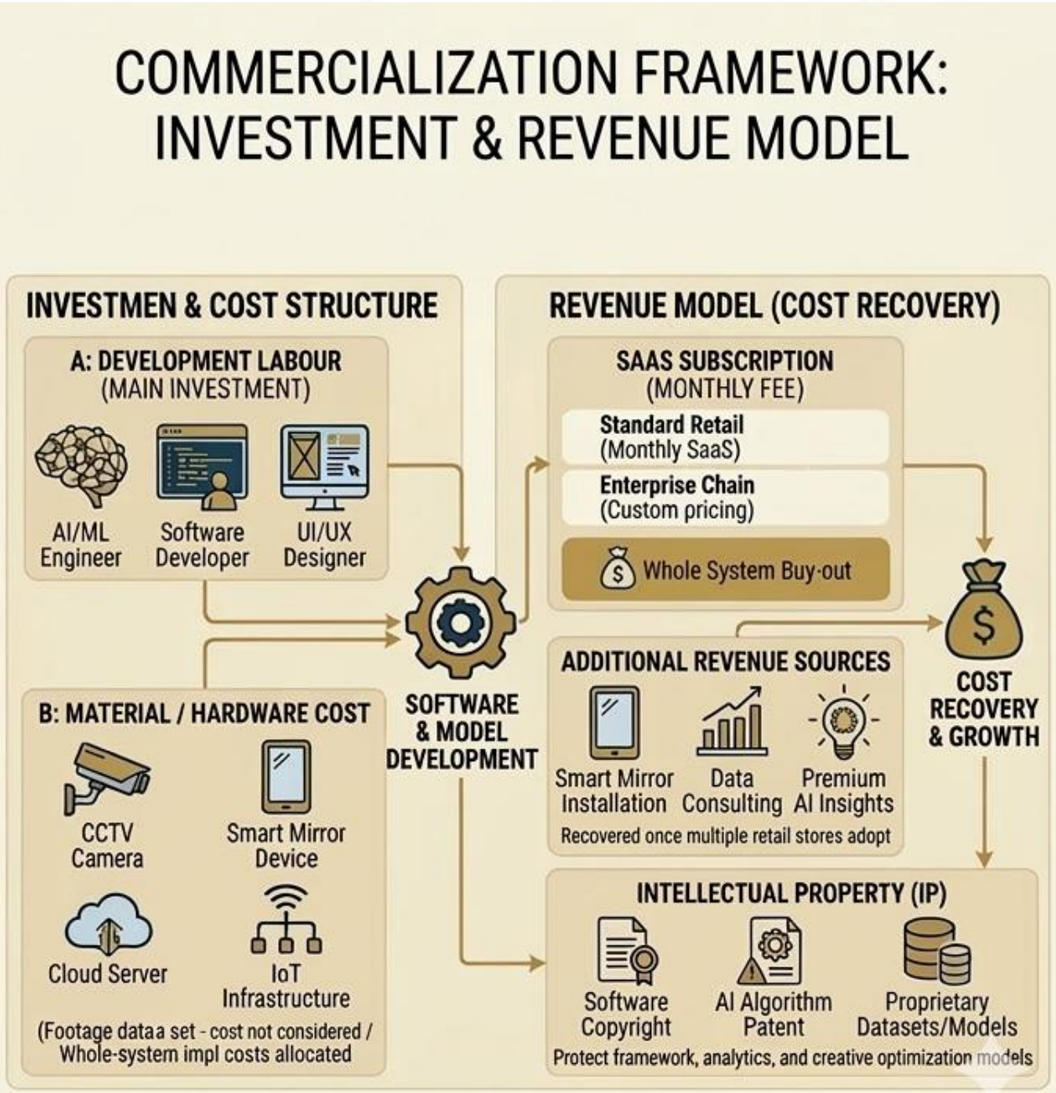
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Gantt Chart



AI-Based Smart Retail Recommendation System for Apparel





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AI-Based Smart Retail Recommendation System for Apparel

Item	Category	Estimated Cost (LKR)	Justification
Cloud Compute (AWS/GCP)	Cloud Services	LKR 180,000 / yr	Kafka, Kubernetes cluster, Snowflake storage for prototype
ML Model Serving (FastAPI)	Infrastructure	LKR 60,000 / yr	Containerised model endpoints via ECS/GKE
OpenAI / Claude API Credits	AI Services	LKR 90,000 / yr	LLM copy generation and NLP processing
Power BI Pro License	Software License	LKR 45,000 / yr	Business-facing KPI dashboards
MLflow Tracking Server	Software	LKR 0 (Open Source)	Model versioning and experiment tracking
Development Hardware	Hardware	LKR 200,000 (one-time)	Developer workstations for local model training
Human Effort (1 developer)	Labour	LKR 480,000 / yr	Full-stack AI/data engineer, 12-month estimate
Total Estimated (Year 1)		LKR 1,055,000	Prototype-to-MVP development budget

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